Genetic engineering

Understanding the structure of DNA has enabled scientists to change or “engineer” the genetic material in cells. It is possible to cut out a gene from one organism (the donor) and place it into the DNA of another organism. When this practice was first attempted in the 1970s it was both difficult and time-consuming, but technological advances —such as Clustered Regularly Interspaced Short Palindromic Repeats, or CRISPR, which has been particularly useful—have greatly simplified and accelerated the process. In theory, geneticists can now splice any gene with any other. They have attempted some intriguing combinations, such as the insertion of the gene for producing spider silk into goat DNA so that goats produce milk rich in proteins. Other substances that can be produced by modifying genes are hormones and vaccines.

In gene therapy, a genetically modified vector (often a virus) is used to carry a gene into the DNA of an organism to replace a faulty or unwanted gene.